Practice workbook · Use a spirometer and identify lung volumes on its trace

How a spirometer records breathing, describing its use to measure tidal volume, and recognising tidal volume, vital capacity and residual volume or exercise changes on a trace.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SPIROMETER-TRACE-INTERPRET

1. Question 5 · part b.i · 4 marks

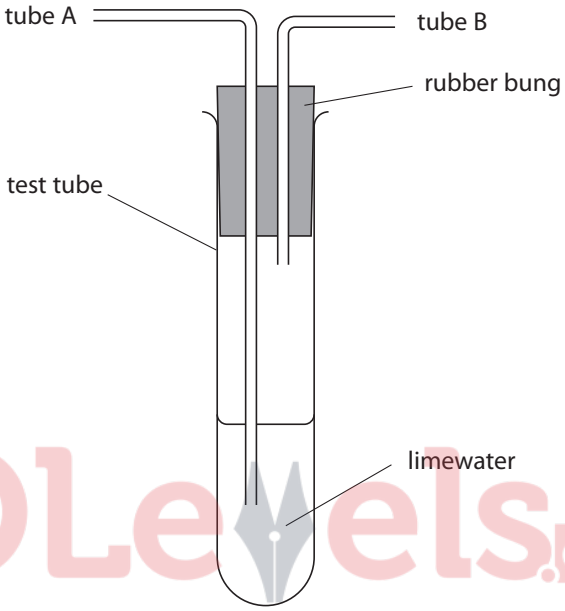
November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

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5 The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.



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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

(4)

1. Question 5 · part b.i · 4 marks

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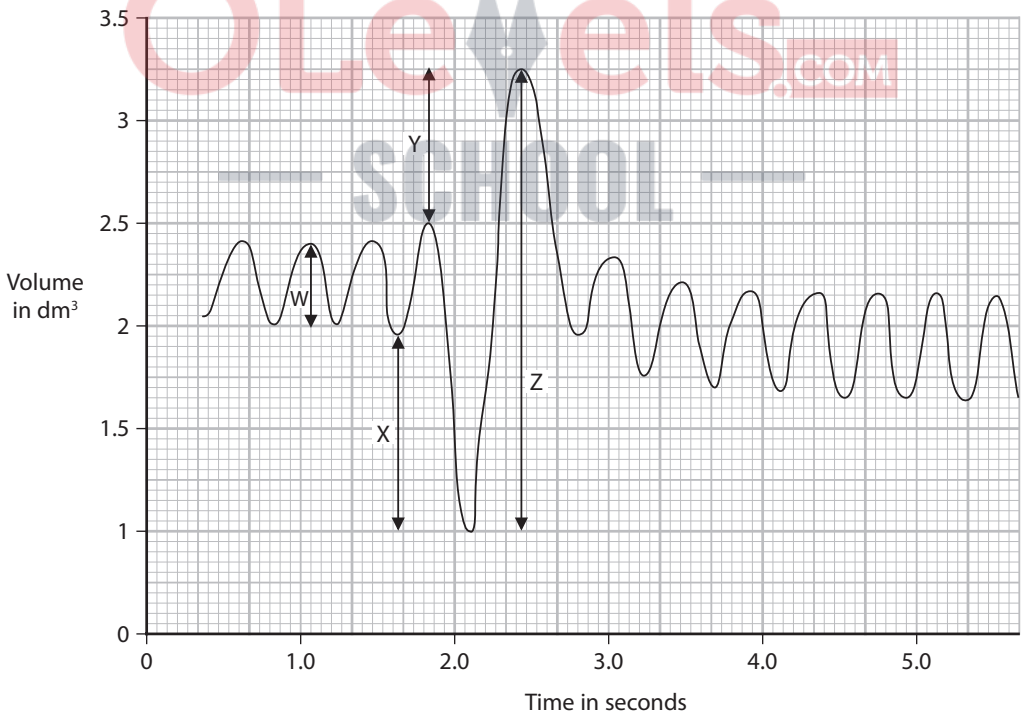
Worked-example retry · Attempt part b.i only.

(ii) Explain the difference expected between the results.

(3)

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.



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1. Question 5 · part b.i · 4 marks

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Worked-example retry · Attempt part b.i only.

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(i) Complete the table using the correct letters from the trace to show the tidal volume, the vital capacity and the volume of air in each case.

(4)

Lung volume	Letter	Volume in dm^3
tidal volume		
vital capacity		

(ii) Explain the pattern of breathing occurring to give lung volume Z.

(2)

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(Total for Question 5 = 13 marks)

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1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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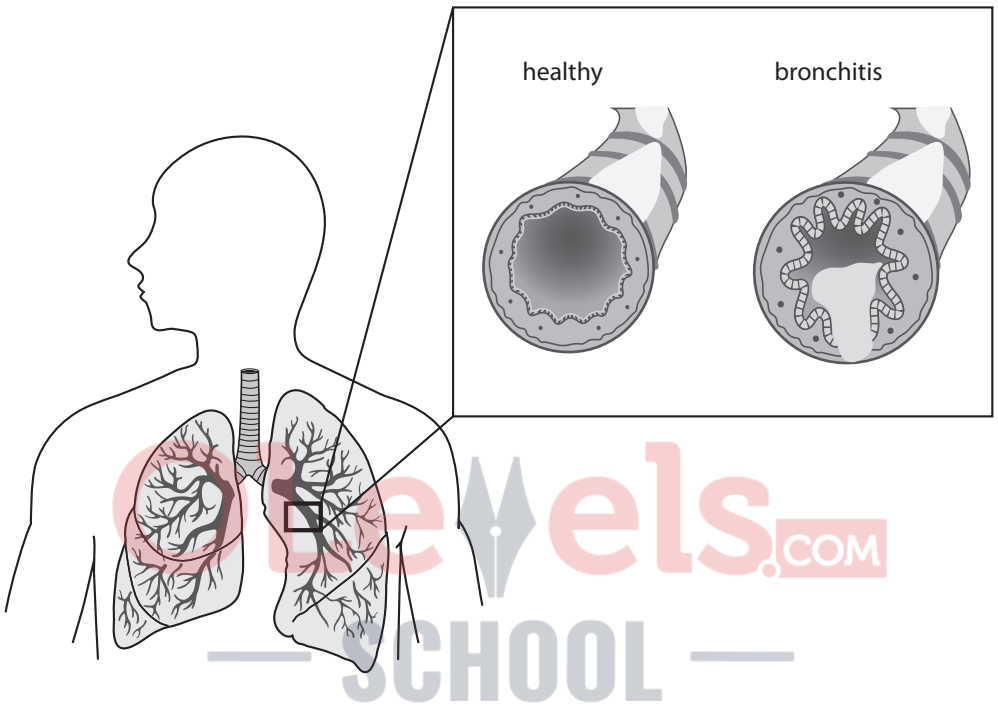
2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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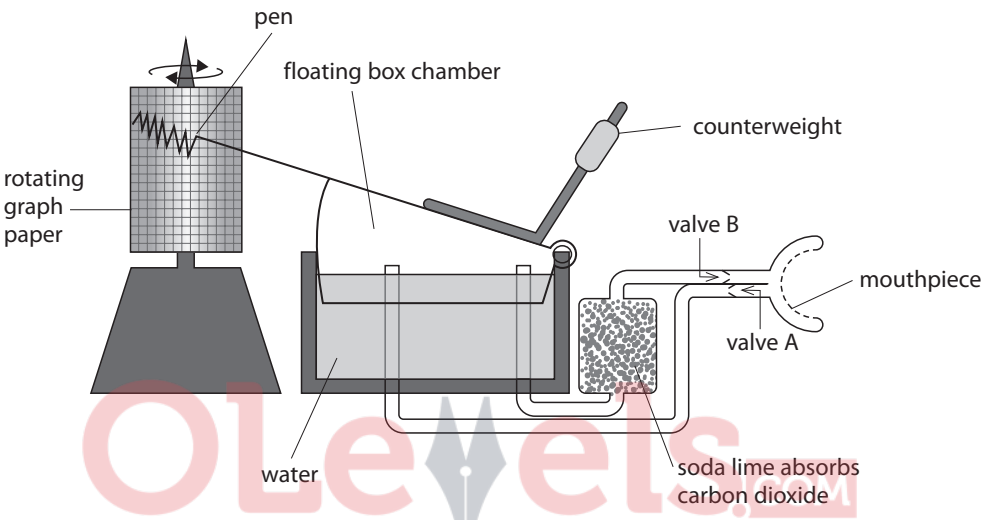
2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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2. Question 6 · part b.i · 4 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

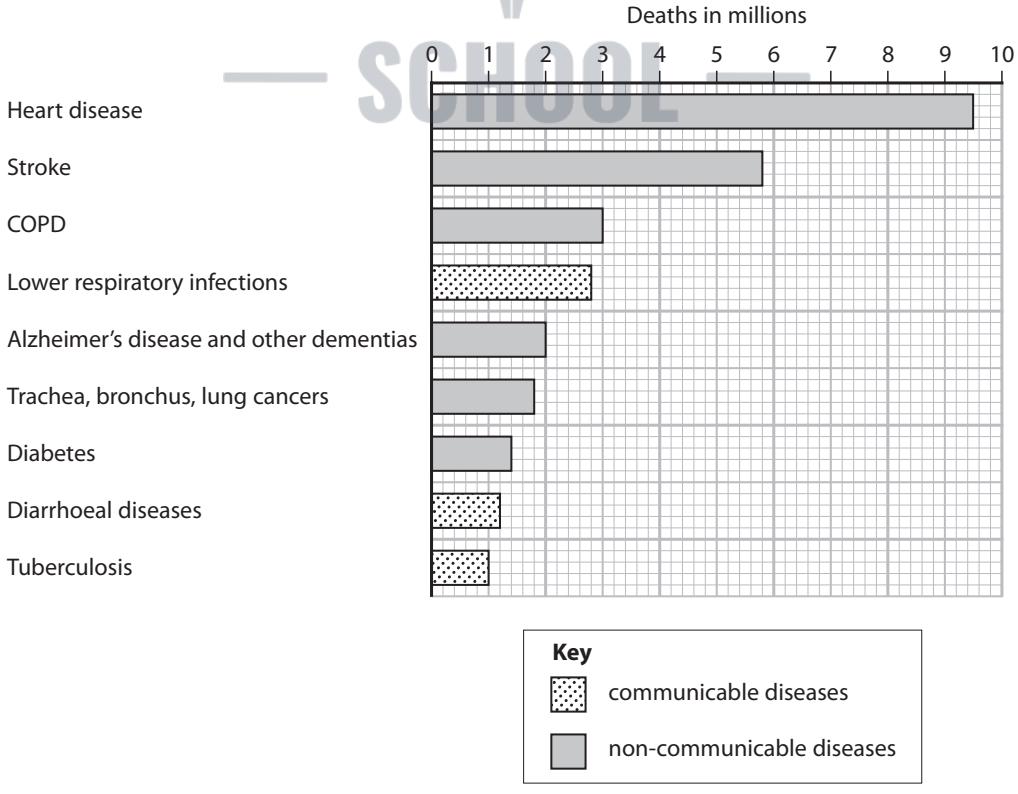
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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2. Question 6 · part b.i · 4 marks · Working and answer

Working and answer

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3. Question 2 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

2 Aerobic respiration is the process that releases energy for use in the body.

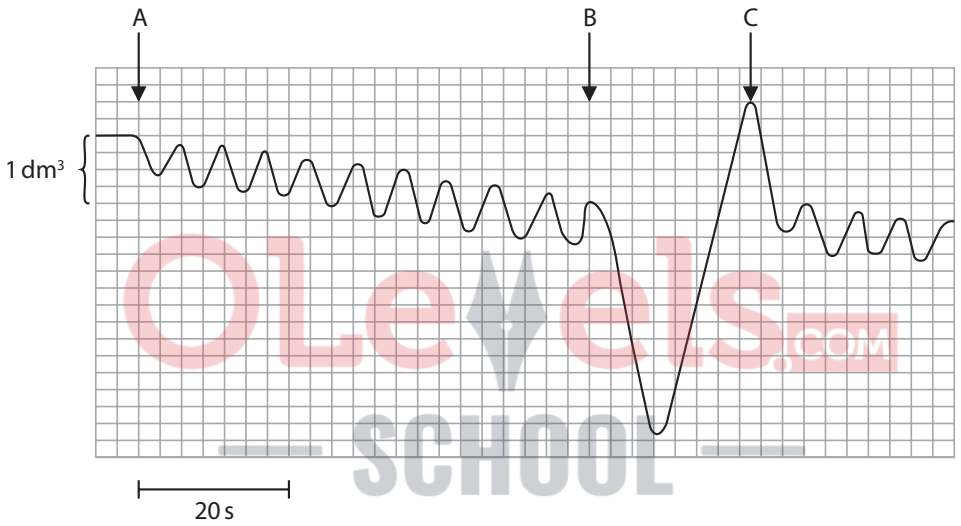
(a) Give the word equation for the process of aerobic respiration.

(4)

..... + ... → ..... + ... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.



(i) Calculate the rate of breathing between points A and B shown on the trace.

(2)

rate = ... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

(3)

difference in volume = ... dm^3

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3. Question 2 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(c) Describe the differences that would be seen in the trace between points A and B if the person had been exercising vigorously before the readings were taken.

(2)

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(Total for Question 2 = 11 marks)

Working and answer

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4. Question 5 · part b.ii · 2 marks

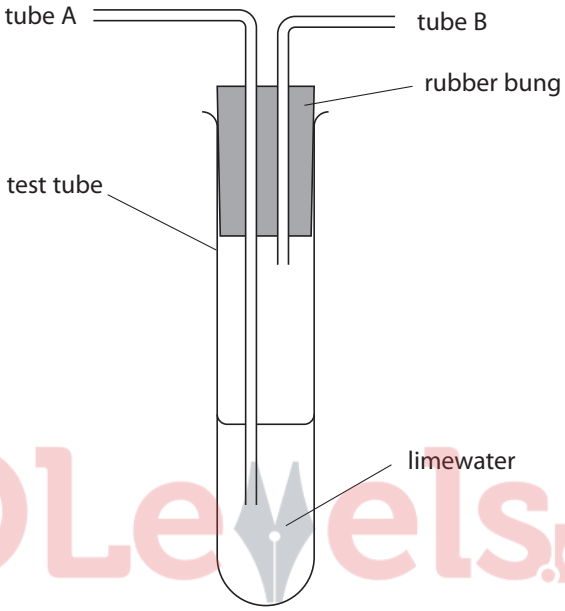
November 2021 Paper 1

Further practice · Attempt part b.ii only.

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5 The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.



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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

(4)

4. Question 5 · part b.ii · 2 marks

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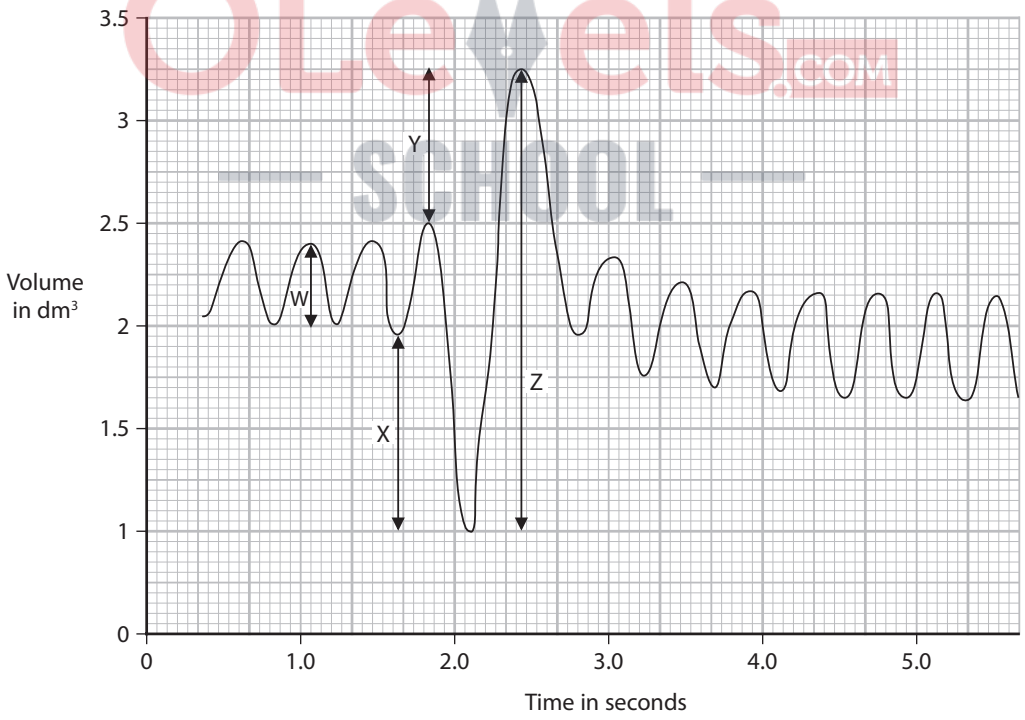
Further practice · Attempt part b.ii only.

(ii) Explain the difference expected between the results.

(3)

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.



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4. Question 5 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(i) Complete the table using the correct letters from the trace to show the tidal volume, the vital capacity and the volume of air in each case.

(4)

Lung volume	Letter	Volume in dm^3
tidal volume		
vital capacity		

(ii) Explain the pattern of breathing occurring to give lung volume Z.

(2)

.....

.....

.....

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(Total for Question 5 = 13 marks)

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Working and answer

5. Question 7 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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5. Question 7 · part c.i · 1 mark

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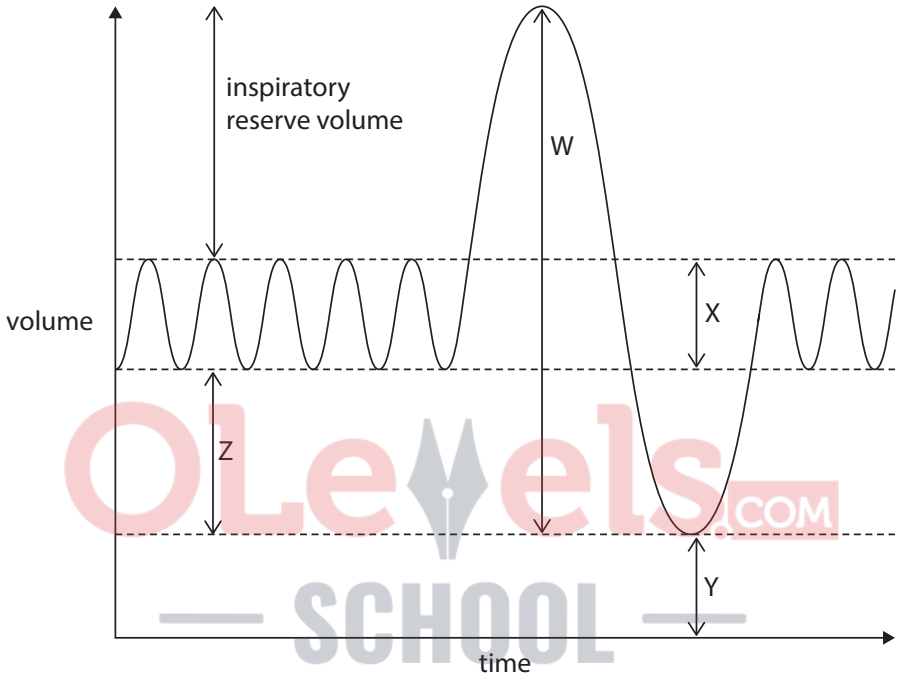
Further practice · Attempt part c.i only.

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(c) The graph shows different lung volumes.



(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

(1)

.....

(ii) Give the name of the lung volume represented by letter Y.

(1)

.....

(iii) Give the name of the lung volume represented by letter W.

(1)

.....

(Total for Question 7 = 9 marks)

5. Question 7 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 7 · part c.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.ii only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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6. Question 7 · part c.ii · 1 mark

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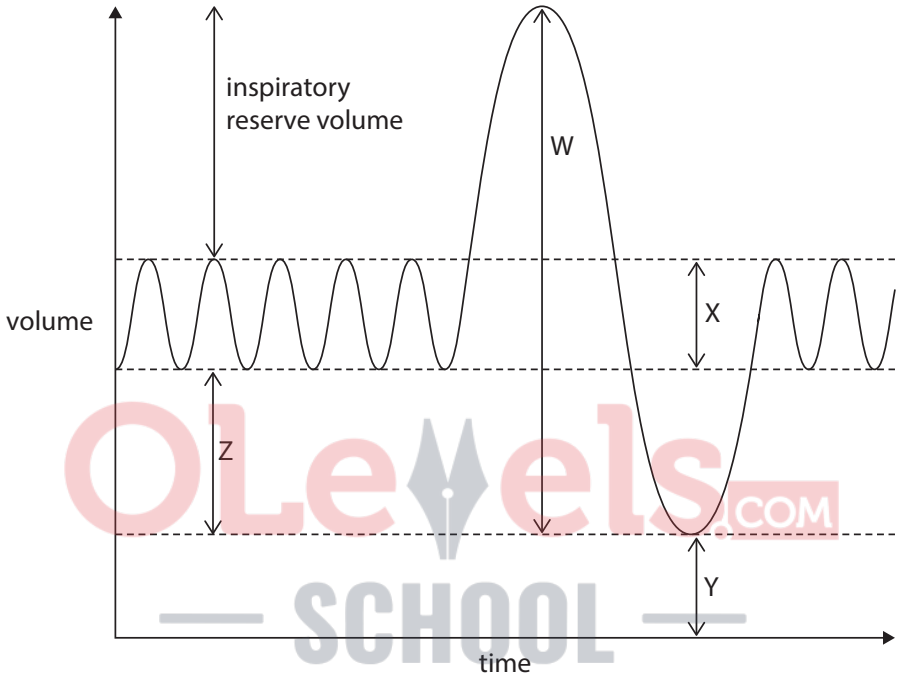
Further practice · Attempt part c.ii only.

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(1)

.....

(iii) Give the name of the lung volume represented by letter W.

(1)

.....

(Total for Question 7 = 9 marks)

6. Question 7 · part c.ii · 1 mark · Working and answer

Working and answer

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7. Question 7 · part c.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.iii only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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7. Question 7 · part c.iii · 1 mark

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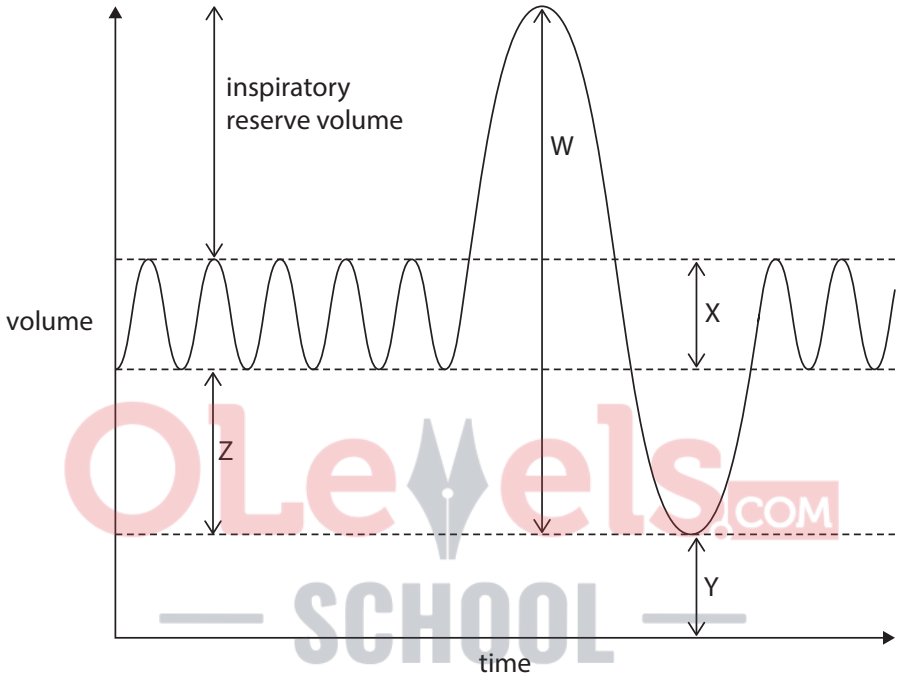
Further practice · Attempt part c.iii only.

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(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

(1)

.....

(ii) Give the name of the lung volume represented by letter Y.

(1)

.....

(iii) Give the name of the lung volume represented by letter W.

(1)

.....

(Total for Question 7 = 9 marks)

7. Question 7 · part c.iii · 1 mark · Working and answer

Working and answer

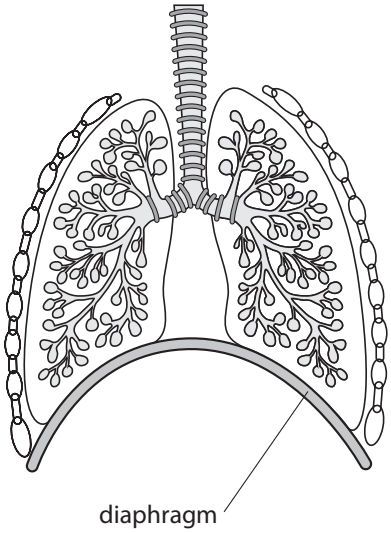
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8. Question 4 · part b.i · 4 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

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2 .....

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3 .....

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4 .....

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8. Question 4 · part b.i · 4 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

(4)

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 cm^3.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

(2)

vital capacity of a person with severe asthma = ... cm^3

(Total for Question 4 = 13 marks)

8. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

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